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ORIGINAL RESEARCH

Heart rate variability as a predictive biomarker of thermal comfort

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Abstract Thermal comfort is an assessment of one's satisfaction with the surroundings; yet, most mechanisms that are used to provide thermal comfort are based on approaches that preclude physiological, psychological and personal psychophysics that are precursors to thermal comfort. This leads to many people feeling either cold or hot in an environment that was supposed to be thermally comfortable to most users. To address this problem, this paper proposes to use heart rate variability (HRV) as an alternative indicator of thermal comfort status. Since HRV is linked to homeostasis, we conjectured that people's thermal comfort could be more accurately estimated based on their heart rate variability (HRV). To test our hypothesis, we analyzed statistical, spectral, and nonlinear HRV indices of 17 human subjects doing light office work in a cold, a neutral, and a hot environment. The resulting HRV indices were used as inputs to machine learning classification algorithms. We observed that HRV is distinctively different depending on the thermal environment and that it is possible to reliably predict each subject's thermal state (cold, neutral, and hot) with up to a 93.7% accuracy. The result of this study suggests that it could be possible to design automatic real-time thermal comfort controllers based on people's HRV.

Keywords Heart rate variability · Thermal comfort · Life engineering · Personalized comfort · Human-centered computing

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1 Introduction

Human thermal comfort is primarily regulated using models based on static heat transfer equations. Fanger's Predicted Mean Vote/Predicted Percentage of Dissatisfied (PMV/PPD) model (Fanger 1970) is a noteworthy example of such models. These static thermal models, however, have well-documented limitations (Van Hoof 2008). For example, the PMV/PPD model is based on laboratory experiments on adults in highly controlled thermal chambers for a relatively extended period. Thus, the model intrinsically assumes fit adults who have achieved a steady state with the surrounding environment; thus, it should not be applied to any other group of people without amendments. Also, it assumes prior knowledge of people's clothing and their activity level. However, people's clothing and activities vary from one person to another and depending on the season of the year. Furthermore, it requires rigid and narrow comfort zones, which are never met in practice (de Dear and Brager 1998; Van Hoof 2008). Finally, static models prescribe a fixed year-round set temperature point for all occupants and ignore any seasonal psychological adaptation on thermal comfort (Lin et al. 2011). Adaptive models have been introduced for an efficient thermal comfort provision. Contrary to the former, they recognize the influence of outdoor settings on how humans adapt to the environment (de Dear and Brager 1998) and are based on field studies (de Dear et al. 2013). Moreover, because people prefer a broad range of thermal comfort zones (Humphreys and Hancock 2007; Nastase et al. 2016; de Dear and Brager 1998), adaptive models' thermal ranges are not as limited as that of the static models. Recently, partly due to an increased awareness of global warming and the importance of environmental conservation, new innovative thermal comfort delivery mechanisms have been proposed. This includes personally controllable air

conditionings, chilled beams, mixed-mode ventilation systems, and a plethora of other exotic methods as recapitulated in de Dear et al. (2013)'s systematic review article. These contemporary systems provide an uneven thermal exposure resulting in a better energy efficiency.

Despite the significant improvement in thermal comfort provision, experts in the fields acknowledge that current thermal comfort regulation mechanisms need overall improvement (Fanger 2001; Croitoru et al. 2015; de Dear 2011; Mahdavi and Kumar 1996; Fountain et al. 1996; Brager et al. 2015; Nicol and Roaf 2017). For example, despite their high energy consumption, the state of the art thermal regulation mechanism are expected to satisfy a mere 80% of the building occupants (Ashrae 2013). Unfortunately, even this inadequate performance is not achieved in practice. In a survey of 215 buildings in North America and Finland, Huizenga et al. (2006) found that only 11% of the buildings achieved the 80% satisfaction rate. On average, 59 ± 16% satisfaction rate was achieved. This is obviously unacceptable as de Dear gloomily decried:

> If the very best that can be achieved in an isothermal, cool, dry and still indoor climate is 'neutral' or 'acceptable' for little more than 80% of a building's occupants at any one time, then the standards that have been set to date leave much to be desired (de Dear 2011, p. 115)

This dearth of thermal comfort delivery is further exposed by an International Facility Management Association (2009)'s survey on complaints in buildings. The survey concluded that many building occupants complain about too hot (91% of the case) and too cold (94%) temperatures. Surprisingly, complaints about other nuisances such as indoor air quality, noise level, limited office space are not as common. Fountain et al. (1996) make a convincing case for why this lack of thermal comfort is expected. They give an example of two people wearing a typical business attire and show that there is no PMV-based set-point temperature that would provide a satisfying thermal comfort for both of them. This is not too surprising since current thermal comfort provision mechanisms aim to provide a satisfactory thermal comfort for an 'average' building occupant. However, since the standard deviation of thermal comfort expectations of a large group of people is high (Ye et al. 2006), in practice, an 'average thermal comfort' is thermally dissatisfying to many people (Fountain et al. 1996). Moreover, most of these thermal comfort provision systems are based on heat transfer and energy conservation principles; thus, innately reflect only the influence of the environment on the person's thermal comfort. However, they neither relate to the complexity of human thermo-regulation nor the adequacy of the provided thermal comfort. For practical reasons, most thermal comfort provision mechanisms assume an all-embracing thermal comfort, i.e., that all people of all gender and age, with different psychological and physiological conditions, have the same thermal comfort expectations. In doing so, they ignore important personal psychophysics that are known to affect the perception of thermal comfort (de Dear and Brager 1998; Karjalainen 2007, 2012; Natsume et al. 1992; Brager and de Dear 1998). Another concern is that these systems make a tacit assumption that people crave for thermal neutrality. A thermal neutral environment is described as an environment in which people do not wish to be warmer or cooler (Mahdavi and Kumar 1996). In reality, however, people tend to prefer non-neutral conditions (Humphreys and Hancock 2007; Nastase et al. 2016; Brager and de Dear 1998; Jones 2002). Ergo, achieving an outright thermal neutrality seem to be meretricious as it does not necessarily reflect neither thermal satisfaction nor thermal pleasure (de Dear 2011). In their articles, de Dear and his coauthors (de Dear 2009, 2011; Parkinson and de Dear 2017, 2016; Parkinson et al. 2016; Parkinson and de Dear 2015) highlighted numerous shortcomings of the one-size-fits-all static thermal settings advocated by most thermal regulation mechanisms. Instead, they proposed *alliesthesia* which is a non-steady state thermal pleasure achieved by varying thermal sensation. It is based on the observation that someone in warm conditions is thermally pleased by cold temperature while the opposite applies to a cold person.

In this paper, we propose to use heart rate variability (HRV) to discretely classify people's thermal comfort state. HRV reflects the time variation between two successive heartbeats. Heartbeats intervals (also referred to as the R–R intervals) are not periodic. Rather, the duration between two consecutive R–R intervals varies from one heartbeat to another (Fig. 1). However, this variation is not random; instead, it changes depending on complex extrinsic protocols imposed on the heart (Acharya et al. 2006). Equally important, HRV is linked to homeostasis (Riganello et al. 2012), i.e., the human body's ability to uphold optimum conditions despite changing external stimuli. For humans, the core temperature is regulated by the brain's hypothalamus that control various mechanism to increase or reduce energy production to restore the core temperature (Hammel and Pierce 1968). Homeostasis is a result of complementary actions of the parasympathetic nervous system (PNS) and the sympathetic nervous system (SNS). The two systems have an antagonistic effect on the heart rate: the SNS increases the heart rate while the PNS has an adverse effect (Robinson et al. 1966; Task Force of The European Society 1996).

Since thermal comfort is, by definition, a personal subjective assessment of the satisfaction of the mind with the thermal environment (Ashrae 2013) and given that thermal changes in environment affect homeostasis (Riganello et al. 2012), which in turn affect HRV (Berntson et al. 1997), our hypothesis is that thermal comfort state could be more

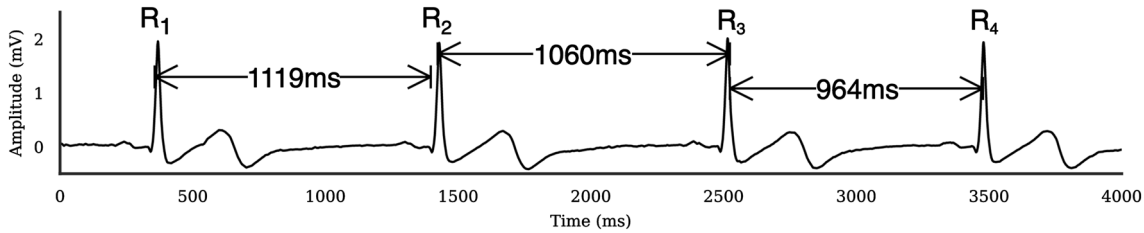

Fig. 1 Heart rate variability: a time variation of consecutive heartbeats (RR intervals)

accurately predicted based on the persons HRV. The main long-term objective our research is to assess the possibility to fully automate indoor air conditioning based on people's HRV. Unlike existing thermal comfort estimation models, using HRV would allow a personalized real-time thermal comfort regulation.

A thorough review of the relevant literature revealed that there is a need for such a study as existing research have limited scopes. For instance, in their various studies, Liu and his colleagues (Liu et al. 2008; Yao et al. 2009) suggested that the sympathetic activity changes noticeably at different temperatures and concluded that heart rate variability might be used as an indicator for thermal comfort. These studies were however only limited to LF/HF, an HRV index whose origin and interpretation is ill-understood and a moot point (Billman 2013; Milicević 2005). Another significant research was conducted at Carnegie Mellon University. This research, however, only focused on the heart rate and skin temperature (Choi 2010; Choi et al. 2012). HRV studies have also been used in sport and exercise medicine. For example, Almeida et al. (2016) demonstrated that HRV indices return to baseline after 60 min of postexercise recovery. This duration is reduced to just 10–20 min when Cold Water Immersion (CWI) is used.

2 Material and method

2.1 Experiment

Seventeen male university students (age 22.35 ± 1.08 years, weight 59.75 ± 7.78 kg, BMI 20.07 ± 2.26) voluntarily participated in the study. All subjects were healthy and did not suffer from any cardiovascular or respiratory illness. Before the experiment, the subjects were requested to abstain from eating, drinking coffee, smoking or doing extensive physical exercise at least two hours before the start of experiment. Aoyama Gakuin University's ethics committee approved the experimental protocols. Subjects wore light clothes (long-sleeved T-shirt and long pants) and were placed inside a climate chamber. During the experiment, the subjects were seated and asked to type randomly selected news articles to simulate office work. The experiments were conducted in three climate chambers whose thermal settings (Table 1) are similar to a cold, a neutral and a hot thermal sensation on the PMV index scale adjusted for cooling effect of elevated air speed (Ashrae 2013; Schiavon et al. 2014). The subjects' electrocardiograms (ECG) were recorded at a 1000 Hz sampling rate and saved for later analysis. The experiment lasted for about 30 min. Right after the experiment, the subjects were requested to rate their thermal comfort sensation on Visual Analog Scale (VAS) consisting of a ten equal intervals line with numerical ranges from 0 (representing the lowest sensation) on the left side to 10 (the highest sensation) on the right side of the line. The subjects were asked to mark a position on the line that best reflects their thermal comfort level, a lower position number indicating a thermal dissatisfaction with the environment.

Table 1 Climate chamber thermal settings

	Cool	Neutral	Hot
Activity level	1	1	1
Clothing level	1	1	1
Air temperature(°C)	18.0	24.0	30.0
Radiant temperature(°C)	18.0	24.0	30.0
Air speed (m/s)	0.3	0.3	0.3
Humidity (%RH)	50.0	65.0	80.0
PMV index*	−1.79	−0.03	+1.87

*PMV adjusted for the cooling effect of an elevated air speed

2.2 HRV calculation

We developed an *ad hot* HRV analysis software. The software enables, *among other things*, the extraction of temporal, spectral and nonlinear HRV indices either from a raw ECG signal or an R–R interval signal. The extraction of R–R interval signal from a raw ECG is based on the Pan and Tompkins (1985) algorithm and the HRV indices are calculated according to standards and algorithms proposed by the Task Force of The European Society (1996) and its subsequent revision by the *e-Cardiology ESC Working Group and the European Heart Rhythm Association* (2015).

In this study, heuristically-selected statistical, spectral, and nonlinear HRV analysis methods were used to calculate various HRV indices summarized in Table 2.

Statistical time domain HRV indices describe the beat-to-beat variability using statistical techniques. They are easy to compute and self-explanatory. In this group, the RMSSD and pNNx indices are of great interest to the HRV research community. The RMSSD represents the square root of the mean of the sum of difference of successive R–R intervals. It is calculated (Eq. 1) as

$$RMMSD = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1}(RR_{i+1} - RR_i)^2} \tag{1}$$

The pNNx, with $x > 0$, is a family of statistical HRV measures of cardiac vagal tone modulation. It indicates the percentage of R–R consecutive pairs that differ by x milliseconds (Eq. 2)

$$pNNx = \frac{1}{N-1}\sum_{i=1}^{N}\left(\left|R_i - R_{i+1}\right| > x\right) \tag{2}$$

Unlike descriptive statistical methods, spectral HRV analysis methods decompose the heartbeat variability into its fundamental frequency components; thus, provide a greater understanding of the fluctuation of the heartbeats. Spectral methods provide an assessment of the interplay between the autonomic nervous system and the heart (Berntson et al. 1997). There exist many approaches to estimating HRV spectra components. Notably, the Fast Fourier Transform (FFT) and the autoregressive (AR) modeling-based techniques are commonly used. However, the two methods assume stationary input signals (in other words, signals whose mean and standard deviation are time-invariant). Therefore, for HRV studies, they should be used with caution as the R–R wave signal is not stationary and is naturally irregularly sampled. In this study, the Lomb-Scargle Periodogram (LSP) (Lomb 1976) was instead preferred. Contrary to most other spectral estimation methods, LSP is suitable to non-stationary and irregularly sampled signals such the R–R wave and it is recommended as the most appropriate method for HRV spectral analysis (Laguna et al. 1998; Moody 1993; Fonseca et al. 2013). A normalized LSP power of a signal X of N data points sampled at time t_i and with $\overline{X}$ as its mean is estimated as (Eq. 3)

$$P(w) = \left\{ \begin{array}{l} \frac{\left[\sum_{i=1}^{N}(X_i - \overline{X}cos[w(t_i-\tau)])\right]^2}{2\sigma^2\sum_{i=1}^{N}cos^2[w(t_i-\tau)]} + \cdots \\ + \cdots \frac{\left[\sum_{i=1}^{N}(X_i - \overline{X}sin[w(t_i-\tau)])\right]^2}{2\sigma^2\sum_{i=1}^{N}sin^2[w(t_i-\tau)]} \end{array} \right\} \tag{3}$$

with w and σ^2 the angular frequency and variance respectively, and the time constant τ defined as in equation 4.

$$\tau = \frac{1}{2w}tan^{-1}\left(\frac{\sum_{i=1}^{N}sin(2wt_i)}{\sum_{i=1}^{N}cos(2wt_i)}\right) \tag{4}$$

In HRV studies, four spectral bands are normally distinguished: the ultra-low frequency [ULF (0–0.0033 Hz)], the very low frequency [VLF (0.0033–0.04 Hz)], the low frequency [LF (0.04–0.15Hz)], and the high frequencies [HF (0.15–0.4 Hz)]. The spectral power of each band is estimated by calculating the area under the spectral density function over its frequency band. Lastly, physiological processes are characterized by phenomena that cannot be measured either by statistical or spectral methods (West

Table 2 Short description of the selected HRV indice

Time domain HRV indice	
Mean RR	Average of all RR intervals
RMSSD	Square root of the mean of the sum of difference of successive RR intervals
SDSD	Standard deviation of difference between adjacent RR intervals
pNNx	Percentage of RR pairs that differ by x milliseconds in the entire recording
Spectral HRV indice	
TP	Total spectral power (0–0.4 Hz)
VLF	Spectral power in very low range frequencies (0.003–0.04 Hz)
LF	Spectral power in low range frequencies (0.04–0.15 Hz)
HF	Spectral power in high range frequencies (0.15 Hz)
LF/HF	Ration between LF and HF power
Nonlinear HRV indice	
$DFA(\alpha 1)$	Short-term fluctuations of detrended fluctuation analysis
$DFA(\alpha 2)$	Long-term fluctuations of detrended fluctuation analysis
$SD1$, $SD2$	Short and long-term variability of the Poincaré plot
SampEn	Sample entropy—a measure of complexity

2010; Sassi et al. 2015; Hardstone et al. 2012). Therefore, nonlinear HRV analysis methods have been used to account for any nonlinear and multi-fractal dynamics in heartbeats variation. A systematic review of such methods can be found in the recent literature (e.g., Sunkaria (2011); Sassi et al. (2015)). In this study, three nonlinear HRV indices were used: the Poincaré plot (Piskorski and Guzik 2007; Brennan et al. 2001), the sample entropy (Lake et al. 2002; Richman and Moorman 2000) and the Detrended Fluctuation Analysis (DFA) (Peng et al. 1995; Willson et al. 2002; Echeverría et al. 2003). Maestri et al. (2007) have shown that these three nonlinear HRV indices are the most reliable for HRV analysis. A Poincaré plot is a diagram in which an RR_i point is plotted against its adjacent RR_{i+1} point. Thus, for an R–R signal RR_1, RR_2, RR_3, RR_N, a Poincaré plot consists of points (X, Y) described (Fig. 5) as,

$$X = RR_1, RR_2, RR_3, \dots RR_{N-1}, \quad Y = RR_2, RR_3, RR_4, \dots RR_N \tag{5}$$

A Poincaré plot has two main descriptors: a short ($SD1$) and long ($SD2$) term descriptor defined in Eqs. 6 and 7.

$$SD1 = \sqrt{var(x_1)} \tag{6}$$

$$SD2 = \sqrt{var(x_2)} \tag{7}$$

with $var(x)$ the variance of x and x_1 *and* x_2 defined as in Eq. 8

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} cos(\pi/4) & -sin(\pi/4) \\ sin(\pi/4) & cos(\pi/4) \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} \tag{8}$$

The $SD1$ and $SD2$ are respectively reported to represent the short term and long term variability. The $SD1$ is believed to indicate the parasympathetic activity while the $SD2$ index reflects the overall HRV variability (Acharya et al. 2006). The sample entropy is another important method used to quantify the predictability of a signal. The entropy itself is a measure of an average information content of a signal (Shannon 1948). The sample entropy uses conditional probability to measure the likelihood that two sequences of length m (that match each other with a tolerance r) will still match if a new sample is added (Humeau-Heurtier 2015). The sample entropy of a signal X is defined as

$$SampEn(m, r) = -\ln\left[\frac{A^m(r)}{B^m(r)}\right] \tag{9}$$

where $A^m(r)$ is the probability that the two sequences will match if a new sample is added to the sequence (thus, the signal is $m + 1$ points) and $B^m(r)$, the probability that the two sequences (of length m) will match. Thus, the sample entropy measures the variability of signal when an extra sample is added. A low entropy signal is more predictable than a higher entropy signal. The sample entropy, unlike some other entropy measures, is much robust against noisy biomedical data such the R–R signal (Richman and Moorman 2000). It is important to note however that its algorithm is influenced by its parameters m and r. Accordingly, the two parameters should be carefully selected (Yentes et al. 2013). In this study, our algorithm uses $m = 2$ and $r = 0.2SDRR$. The Detrended Fluctuation Analysis (DFA) is yet another nonlinear HRV analysis method. It repeatedly assesses the signal at different times. Thus, in the case of physiology signal, DFA allows distinguishing short-term perturbations $DFA(\alpha 1)$ from perturbation dues to long lasting perturbations $DFA(\alpha 1)$ which are due to, e.g., the body's internal functionality. In our study, the algorithm is implemented according to an algorithm described by Voss et al. (2012). The algorithm consists of the following four major steps. First, the $R - R$ signal is integrated (Eq. 10).

$$y_{int}(k) = \sum_{i=1}^{k}(RR_i - \overline{RR}), \quad k = 1, 2, \cdots N \tag{10}$$

In the above equation, $\overline{RR}$ is the average of the signal. Second, the integrated signal is split into N non-overlapping equal length segments of length n. Third, each segment is 'detrended' by removing a polynomial fit calculated using a least-square regression. Lastly, the Root Mean Square (RMS) fluctuation value $F(n)$ of the signal is computed (Eq. 11)

$$F(n) = \frac{1}{N}\sum_{k=1}^{N}\left(y_{int}(k) - y_{fit}(k)\right)^2 \tag{11}$$

In this study, the DFA's short term exponent $DFA(\alpha 1)$ was computed for a range of n = 4 to 16, while the long-term exponent $DFA(\alpha 2)$ was computed for $n = 16$ to 64 as recommended by Peng et al. (1995).

2.3 Statistical analysis

The normal distribution of each HRV index was tested using a Levene's test. The test revealed that most HRV indices were not normally distributed; thus, a Mann–Whitney–Wilcoxon test for two independent samples was used to verify the statistical significance of HRV parameters in a different environment. This test is a non-parametric test that estimates the probability that a random sample from one group is less than or greater than another randomly selected sample from another group; thus, this test is used to compare two independent groups of data.

2.4 Machine learning classification

Supervised Machine Learning (ML) classification algorithms were used to build models that could be used to predict the comfort status of a person. In this study, Logistic Regression (LR), Linear Discriminant Analysis (LDA), K-Nearest Neighbors (KNN), Decision Tree (DT), Naïve Bayes (NB), Support Vector Machine (SVM), Random Forest (RF), Multi-Layer Perceptron (MLP), Adaptive Boosting (ADABOOST), and Quadratic Discriminant Analysis (QDA) classifiers were trained to predict if a subject is in a cold, neutral, or hot environment based on the subject's HRV indices. Kotsiantis et al. (2006) provide a concise review and algorithmic implementations of the aforementioned ML classifiers. Because there exist many HRV indices, some of which may contain redundant information, we used a Randomized Logistic Regression (RLR) Meinshausen and Bühlmann (2010) to select the most important HRV indices for classifiers. In a nutshell, the RLR algorithm fits a randomly selected subset of the data; then, it combines and analyzes the results from different HRV indice. The HRV indices deemed less significant in classifying the data are rejected while the important one are retained. The RLR method has a twofold advantage: It increases the accuracy of the model and reduces the complexity and computation requirements of the model. The classification accuracy of each ML classifier was estimated using a 10-fold stratified cross-validation.

2.5 HRV indices computation

In this study, we were interested in understanding how a thermal environment affects the evolution heartbeat. Thus, to compute the HRV indices listed in Table 2, for each subject, the HRV indices were first extracted from a 5-min-long ECG window segment. Then, the window segment is shifted by approximately 15 s and new HRV indices are calculated. This process is repeated until the end of the entire ECG recording. In this paper, since different HRV indices have different numerical ranges, the result is reported on a normalized scale between 0 and 1.

3 Results

In general, subjects reported feeling cool or cold in the cold chamber, comfortable in the neutral thermal chamber and warm or hot in a hot thermal chamber. However, as shown in Fig. 2, different subjects reported a different level of thermal sensations for the same thermal environment.

A comparison of HRV in the three different environments reveals that HRV is distinctly different from one environment to another. A thorough analysis of the various HRV indices uncovers four important observations. First, in general, the $DFA(\alpha 1)$ is disproportionately highest in the hot environment and mostly lowest in the cold environment as illustrated in Fig. 3 and summarized in Table 3.

However, this trend is not unanimous to all subjects. A few subjects violate this general trend as indicated in Fig. 4.

Second, the Mean RR, the VLF and the sample entropy HRV indices are noticeably highest in the cold environment and lowest in the hot environment as illustrated in Fig. 5 and recapitulated in Table 3.

However, unlike the $DFA(\alpha 1)$, the separation between the three environments is clearer and persisted during the whole duration of the experiment. As with the $DFA(\alpha 1)$, some subjects were exceptional to this trend (Fig. 6).

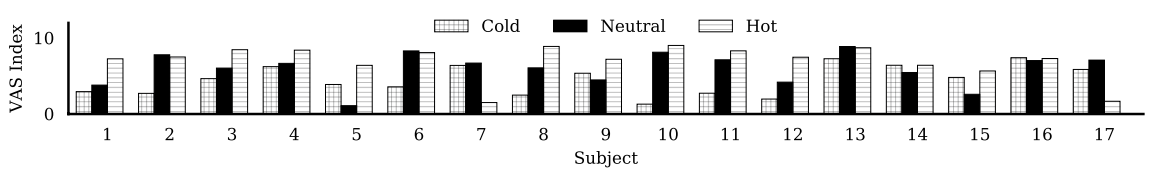

Fig. 2 Self-reported thermal sensation indicating that subjects have different thermal sensation for the same environment

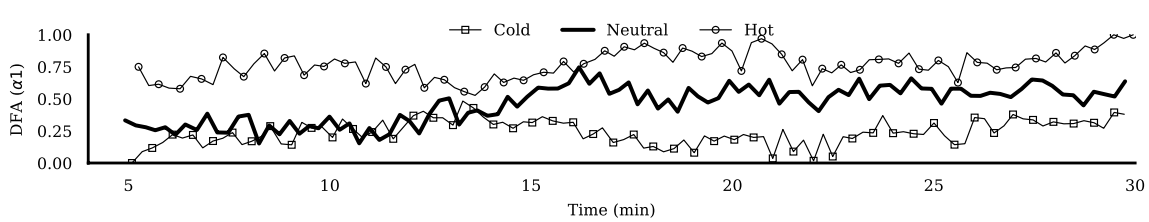

Fig. 3 Normalized mean $DFA(\alpha 1)$ indicating that the $DFA(\alpha 1)$ is consistently highest in the hot environment and mostly lowest in the cold environment

Table 3 Comparison of HRV indices values in different thermal settings

	Cold (mean ± std)	Neutral (mean ± std)	Hot (mean ± std)
Mean RR (ms)	824.47 ± 91.42	795.13 ± 79.55	768.69 ± 83.52
SDRR (ms)	54.25 ± 19.33	55.56 ± 23.51	44.34 ± 13.32
RMSSD (ms)	43.37 ± 20.57	42.73 ± 30.74	31.00 ± 14.15
SDSD (ms)	43.36 ± 20.57	42.73 ± 30.74	31.00 ± 14.15
pNN25 (%)	49.49 ± 18.11	43.77 ± 16.95	35.84 ± 20.79
VLF (ms^2)	4442.82 ± 483.55	4291.14 ± 418.26	4141.95 ± 442.64
LF (ms^2)	1800.04 ± 206.71	1777.18 ± 203.12	1723.08 ± 213.18
HF (ms^2)	2554.58 ± 491.47	2607.48 ± 545.67	2482.16 ± 528.45
TP (ms^2)	8834.73 ± 987.00	8713.29 ± 925.75	8383.47 ± 928.97
LF/HF	0.72 ± 0.09	0.70 ± 0.11	0.71 ± 0.11
Sample entropy	1.72 ± 0.22	1.58 ± 0.30	1.62 ± 0.24
DFA ($\alpha 1$)	1.10 ± 0.21	1.15 ± 0.24	1.22 ± 0.23
SD1 (ms)	76.47 ± 27.28	78.33 ± 33.19	62.50 ± 18.79
SD2 (ms)	69.71 ± 24.53	71.53 ± 27.50	58.30 ± 17.18
SD1/SD2	1.10 ± 0.06	1.09 ± 0.07	1.07 ± 0.05

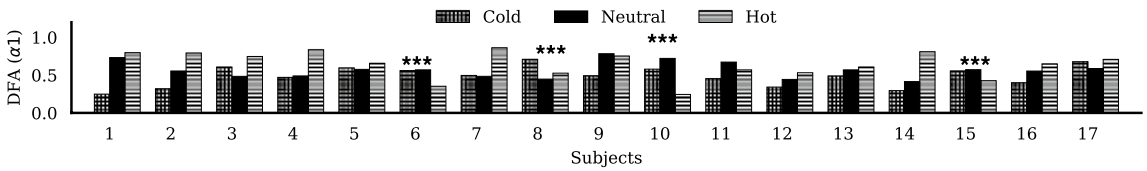

Fig. 4 Inter-subject comparison of a normalized indicating that, for most subjects, the *DFA*($\alpha 1$) is highest in the hot environment and lowest in the cold environment. The *three star sign* indicates irregularity in this trend, i.e., where the *DFA*($\alpha 1$) was higher in the cold than in the hot environment

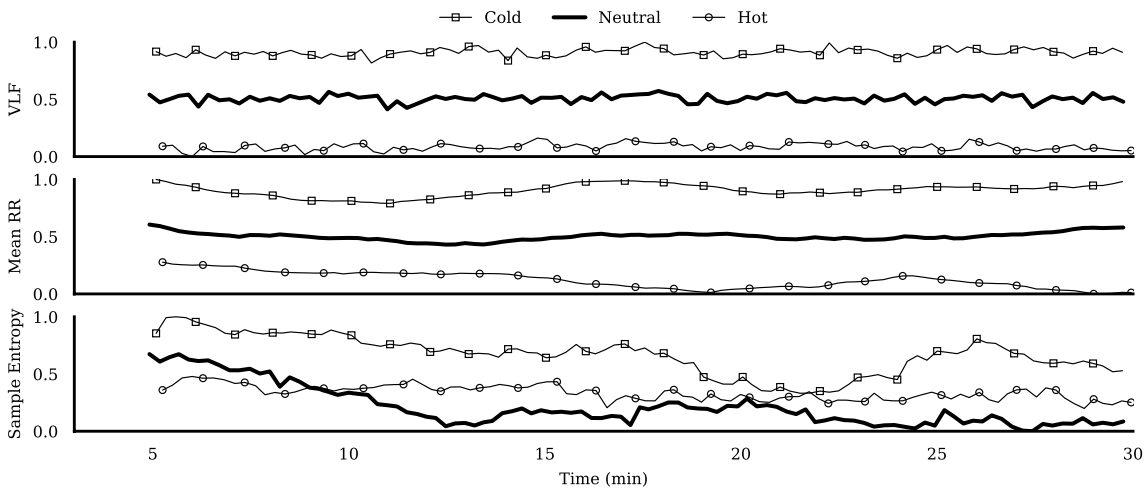

Fig. 5 The normalized mean of HRV indices that are highest in the cold environment

Third, the pNNx family of HRV indices are also highest in the cold environment and lowest in the hot environment (Fig. 7).

Moreover, as shown in Fig. 8, this difference is much more pronounced for an R–R increment roughly between 20 and 40 ms and gradually decreases outside this boundary.

Fourth, for other HRV indices, the separation between the hot and cold environment is not as clear over the whole duration of the experiment. Instead, as shown in Fig. 9, there is no clear distinction between the three thermal environments as the plots get mixed up with time. This is the case for spectral indices (TP, LF, LF/HF and HF), the Poincare

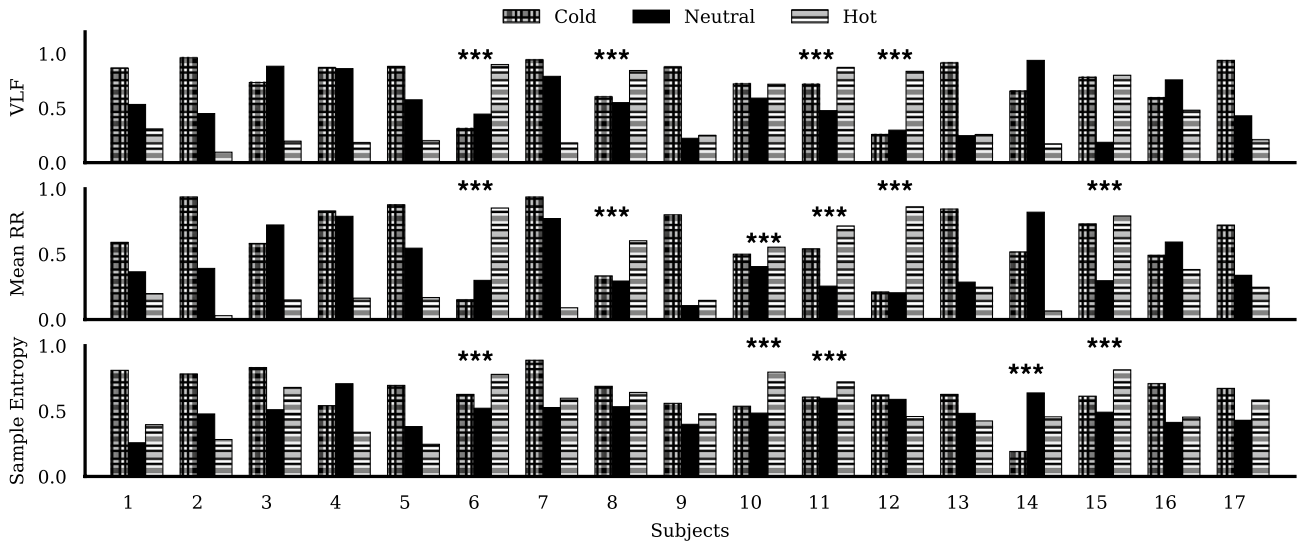

Fig. 6 Inter-subject comparison of normalized HRV indices highest in the cold environment. The *three stars* (***) indicates irregularity in this trend, i.e., where the HRV indices were higher in the hot than in the cold environment

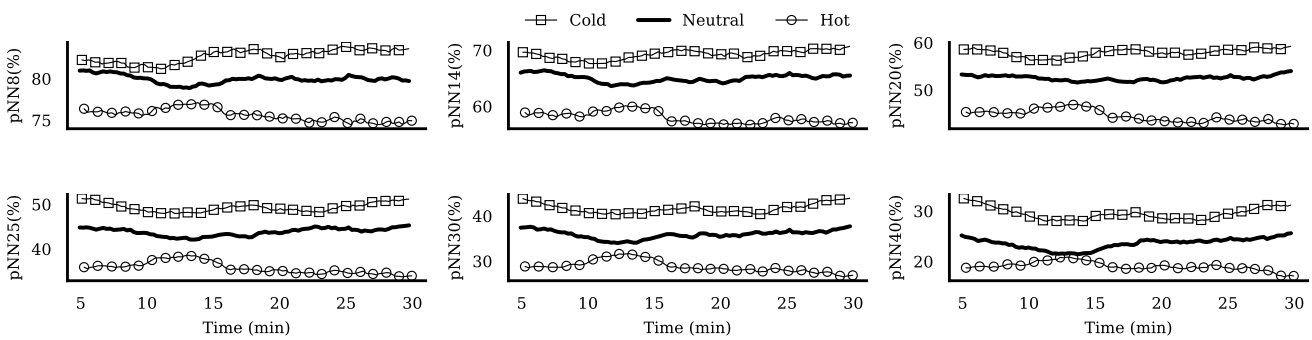

Fig. 7 Time variation of some pNNx HRV indices indicating that pNNx is highest in the cold environment and lowest in the hot environment

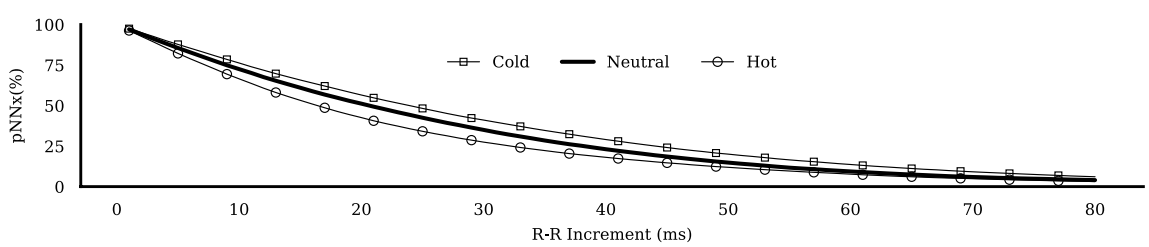

Fig. 8 pNNx variation with an increase in an R–R increment x. Note that the separation between the hot and cold environment is highest between 20 and 40 ms and gradually decreases outside this boundary

plots based indice, and some statistical HRV indices such as the RMSSD, the SDRR, and the SDSD.

A Randomized Logistic Regression (RLR) feature selection algorithm was applied to obtain the most relevant features for thermal comfort prediction. The algorithm selected the following six HRV indices: Mean RR, the RMSSD, the SDSD, the pNN25, the VLF and the sample entropy. These HRV indices were used to make ML models that were afterward used to predict the thermal environment of a subject given his HRV indice. The highest accuracy was achieved by a Support Vector Machine (SVM) classifier at 93.7% (Table 4).

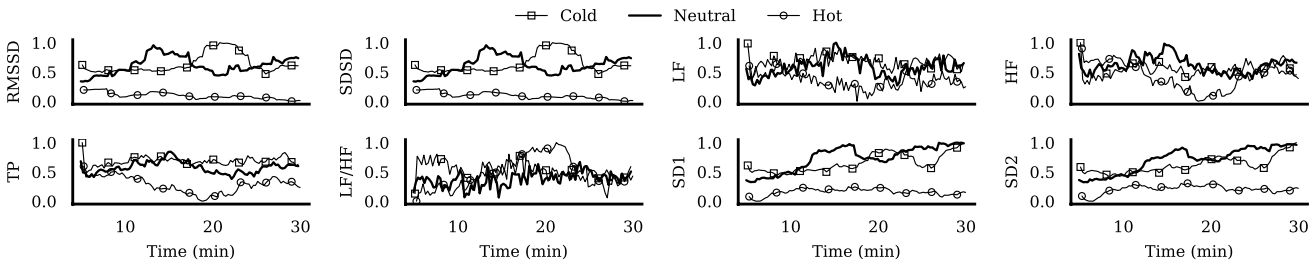

Fig. 9 HRV indices whose temporal variation does not distinguish the three environments

Table 4 Cross-validation classification accuracy for each subject

Subject	Machine learning classifier accuracy (%)									
	LR	LDA	KNN	DT	NB	SVM	RF	MLP	ADBOOST	QDA
1	93.18	96.89	93.86	91.83	94.76	96.59	95.26	96.3	85.62	95.98
2	100	100	100	100	100	100	100	100	100	100
3	90.72	94.87	92.36	92.25	86.76	96.79	96.39	94.88	93.13	89.75
4	79.66	81.19	82.18	86	77.52	90.5	88.85	88.55	86.52	85.17
5	100	100	100	99.02	100	100	99.33	100	99.02	99.68
6	90.6	99.02	98.69	97.69	99.02	99.35	95.83	99.67	96.45	97.7
7	91.79	99.67	99.67	100	99.33	100	100	100	100	99.33
8	80.49	79.89	84.19	75.78	71.23	80.66	84.62	81.43	71.85	77.42
9	77.38	71.85	79.16	82.64	75.4	81.55	80.3	75.63	75.01	78.06
10	92.68	95.42	91.69	91.36	89	96.09	93.91	95.45	89.7	98.36
11	70.25	74.55	74.74	77.49	66.68	82.99	84.39	81.62	66.24	74.63
12	95.71	95.38	93.85	96.25	93.74	96.28	93.52	95.3	96.55	96.89
13	74.38	73.64	80.72	82.27	75.03	83.98	77.7	80.42	74.52	82.49
14	99.67	100	98.47	98	100	100	100	99.69	100	100
15	99.7	100	100	100	100	100	100	99.09	100	100
16	88.39	85.81	91.71	86.68	83.3	93.64	89.01	90.01	68.17	93.39
17	87.61	97.78	93.54	92.63	92.84	95.27	96.82	97.09	97.78	95.83
Average	88.95	90.94	91.46	91.17	88.51	93.75	92.70	92.65	88.27	92.04

4 Discussion

Despite advancement in thermal comfort alleviation technologies and the amount of energy used to provide it, most building occupants are not satisfied with the thermal environment in the building. A prominent survey International Facility Management Association (2009) brought to light this lack of comfort satisfaction and blamed too hot and too cold temperatures as the primary catalysts for the discomfort.This may be due to current thermal regulation systems that mostly ignore personal physiological and psychological precursor to thermal comfort. In this paper, we investigated the possibility to use HRV as an alternative indicator of thermal comfort. The study was conducted in three artificial thermal environments: cold, neutral, and hot on a PMV scale. In general, most subjects reported feeling cool or cold in the cold chamber, comfortable in the neutral environment and warm or hot in a hot environment. However, the level of thermal sensation slightly fluctuated from one subject to another. While there is a possibility that this variation is due to a bias of self-report questionnaires, other researchers (Humphreys and Hancock 2007; Nastase et al. 2016; Brager and de Dear 1998; Jones 2002) believe that different people have different thermal preferences. This discrepancy is due in part to variations in physiological and physiological differences among people. It may also be a result of inter-subjects thermal perception variability. In all cases, however, no subject expressed being hot in a cold environment or being cold in a hot environment; thus, we concluded that the subjects felt cold in a cold environment and hot in a hot environment.

The study found that HRV evidently varies from one environment to another ($p < 1 \times 10^{-4}$). The short-term DFA coefficient is consistently highest in the hot environment and lowest in the cold environment. The DFA is a measure of self-affinity and long-term fractal correlation in the signal

(Peng et al. 1995). In HRV analysis, DFA is known to provide unique information that cannot be obtained from classical HRV analysis methods (Hardstone et al. 2012; Penzel et al. 2003) and is used to for example to distinguish healthy subjects from sick ones (Rodriguez et al. 2007). It is worthy to note that, while there is a clear separation between the hot and cold environments, for the $DFA(\alpha 1)$, the split-up between the cold and neutral environment is not as strong (Fig. 3).

Other indices such as Mean RR, the VLF, and the sample entropy are highest in the cold environment and lowest in the hot environment. The VLF and the sample entropy HRV indices are especially interesting for our study. Unlike other HRV indices thus far, VLF is a result of analyzing the spectral components of the variability of successive R–R intervals. Thus, it reflects best the action of the nervous system on the heart rate variability. Moreover, some studies have linked VLF to thermal regulation (Fleisher et al. 1996; Thayer et al. 1997); thus, it could be a good indicator of thermal sensation. As for the sample entropy, it is highest in the cold and hot environments and lowest in the neutral environment (Fig. 5). The sample entropy is a measure of unpredictability. In our case, it seems that the sample entropy is highest in non-thermally comfortable environments (hot and cold) and lowest in a comfortable environment (neutral). This suggests that, in the absence of thermal comfort, the heartbeats are more unpredictable (thus, the increase in entropy). This increase of the sample entropy HRV index suggests that some new external catalyst influences the heart's functionality, and in this study, that external influence can only be the change in thermal environmental conditions. This observation seems to be further confirmed by analyzing the balance between the parasympathetic nervous system (PNS) and the sympathetic nervous system (SNS) in the three thermal environments. The parasympathetic system is responsible for the 'rest and digest' activities to restore the balance of the body systems. In case a of any perceived stressful situation, these activities are momentary minimized yield to the 'fight or flight' sympathetic nervous system. As shown in Table 3, HF is highest the neutral condition and is lowest in the two uncomfortable thermal conditions (cold and hot). Since HF is known as an indicator of the cardiac parasympathetic nerve activity, it is safe to conclude that, in a non-thermal comfortable environment, the body increases its parasympathetic activity to the detriment of the parasympathetic activity. This results in the increase of the LF/HF ratio which is believed to correlated with the cardiac sympatho-vagal balance (Table 3). The pNNx family of HRV measure is also steadily highest in the cold environment and lowest in the hot environment. As shown in Fig. 8, the difference is largest for an R–R increment between 20 and 40 ms. For our study, the pNNx is also of a particular interest. Unlike other HRV indices, it is less computation intensive to calculate; thus, could be used to estimate thermal comfort using less powerful embedded micro-controllers.

Finally, the study investigated the possibility to automatically predict if people feel cold or hot based on their HRV. Since thermal sensation is linked to homeostasis which in turn affect HRV, we posit that HRV could be used to predict when people are cold, comfort, or hot. We assumed that, since the subjects' self-report survey indicated that they were indeed cold in a cold environment and hot in a hot environment, our prediction based on their HRV would closely reflect their thermal sensation. Our machine learning-based classification model achieved a 93.7% accuracy. Although the highest classification accuracies were achieved by complex models, much simpler, and less computation intensive models such as the NB classifiers provide adequate accuracy (Table 4). The results of this study lead us to believe, as we initially surmised, that it is likely possible to design automated thermal controllers that predict people's comfort state based on their HRV.

5 Conclusion

In our quest to design a more efficient thermal comfort controller, we proposed HRV as a biomarker for thermal comfort. The following findings emerged from this study:

1. HRV is distinctively altered depending on the thermal environment. Some HRV indices were highest in the hot environment [notably the $DFA(\alpha 1)$], and others were highest in the cold environment (the VLF, mean RR, the sample entropy, and the pNNx). For these indices, the difference between the hot and the cold environment was steady throughout the experiment.
2. The sample entropy was highest in the uncomfortable environments (the cold and the hot). Since a high entropy indicates an increase in unpredictability in a signal, this confirms our assumption that a change in thermal environment indirectly influences a detectable change in the heart's functionality as the person adapts to the thermal stress. This suggests that the heartbeat is more regular and less complex in thermally comfortable environments and exhibits a more complex pattern in non-thermally comfortable environments.
3. The cardiac sympatho-vagal balance increases in non-thermal comfortable environment (cold and hot) and decreases in thermally comfortable environments.
4. The study pinpointed six HRV indices that are most significant in predicting the thermal state of the subjects. These are: The Mean RR, the RMSSD, the SDSD, the pNN25, the VLF and the sample entropy

5. Using machine learning algorithms, it was possible to reliably predict each subject's thermal state (cold, neutral, and hot) with up to a 93.7% accuracy.

6 Practical implication

The results of this study suggests that it could be possible to design automated real-time thermal comfort controlled based on people's HRV. While we recognized that this is not a panacea for all thermal comfort requirements, we believe that, since non-neutral thermal conditions do not result necessarily in thermal discomfort, it could be possible to design a computerized system that let the indoor temperature drift away from the thermal neutrality and adjust it only if people feel cold or hot. This approach would reduce the energy used for thermal comfort regulation while providing acceptable thermal comfort. In their simulations, Hoyt et al. (2014) demonstrated that it could be possible to save 29% of cooling energy and 27% total HVAC energy by simply expending cooling setpoint by a mere 2.8 °C. Likewise, lowering the heating setpoint by 1.1 °C has a potential to save an average of 34% of heating energy. Furthermore, due to the availability of non-contact ECG sensors and a proliferation of inexpensive wearable devices with built-in photoplethysmography (PPG) sensors (e.g., smart-watches and fitness trackers), this objective could be achieved in a non-invasive manner and at low cost. PPG is a reliable unobtrusive technique to monitor beat-to-beat heart rate variability via arterial oxygen saturation and is known as a good HRV surrogate (Selvaraj et al. 2008; Bolanos et al. 2006).

References

Acharya UR, Joseph KP, Kannathal N, Lim CM, Suri JS (2006) Heart rate variability: a review. Med Biol Eng Comput 44(12):1031–1031. doi:10.1007/s11517-006-0119-0 **(ISSN 01400118)**

Almeida AC, Machado AF, Albuquerque MC, Netto LM, Vanderlei FM, Vanderlei LCM, Junior JN, Pastre CM (2016) The effects of cold water immersion with different dosages (duration and temperature variations) on heart rate variability post-exercise recovery: a randomized controlled trial. J Sci Med Sport 19(8):676–681. doi:10.1016/j.jsams.2015.10.003 **(ISSN 14402440)**

ASHRAE (2013) ASHRAE/ANSI standard 55-2013 thermal environmental conditions for human occupancy. American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Atlanta, GA

Berntson GG, Bigger JT, Eckberg DL, Grossman P, Kaufmann PG, Malik M, Nagaraja HN, Porges SW, Saul JP, Stone PH, van der Molen MW (1997) Heart rate variability: origins, methods, and interpretive caveats. Psychophysiology 34(6):623–648

Billman GE (2013) The LF/HF ratio does not accurately measure cardiac sympatho-vagal balance. Front Physiol. doi:10.3389/fphys.2013.00026**(ISSN 1664042X)**

Bolanos M, Nazeran H, Haltiwanger E (2006) Comparison of heart rate variability signal features derived from electrocardiography and photoplethysmography in healthy individuals. Annu Int Conf IEEE Eng Med Biol Proc. doi:10.1109/IEMBS.2006.260607 **(ISSN 05891019)**

Brager GS, De Dear RJ (1998) Thermal adaptation in the built environment: a literature review. Energy Build 27(1):83–96. doi:10.1016/s0378-7788(97)00053-4 **(ISSN 03787788)**

Brager G, Zhang H, Arens E (2015) Evolving opportunities for providing thermal comfort. Build Res Inf 43(3):274–287

Brennan M, Palaniswami M, Kamen P (2001) Do existing measures of Poincar?? plot geometry reflect nonlinear features of heart rate variability? IEEE Trans Biomed Eng 48(11):1342–1347. doi:10.1109/10.959330 **(ISSN 00189294)**

Choi JH (2010) CoBi: bio-sensing building mechanical system controls for sustainably enhancing individual thermal comfort. PhD Dissertation. Carnegie Mellon University. Retrieved from http://repository.cmu.edu/dissertations/33. Accessed 20 Aug 2017

Choi JH, Loftness V, Lee DW (2012) Investigation of the possibility of the use of heart rate as a human factor for thermal sensation models. Build Environ 50:165–175

Croitoru C, Nastase I, Bode F, Meslem A, Dogeanu A (2015) Thermal comfort models for indoor spaces and vehicles—current capabilities and future perspectives. Renew Sustain Energy Rev 44:304–318. doi:10.1016/j.rser.2014.10.105 **(ISSN 13640321)**

de Dear RJ, Akimoto I, Arens EA, Brager G, Candido C, Cheong KWD, Li B, Nishihara N, Sekhar SC, Tanabe S, Toftum J, Zhang H, Zhu Y (2013) Progress in thermal comfort research over the last twenty years. Indoor Air 23(6):442–461 **(ISSN 09056947)**

de Dear R (2009) The theory of thermal comfort in naturally ventilated indoor environments—the pleasure principle. Int J Vent 8(3):243–250. doi:10.1080/14733315.2009.11683849 **(ISSN 1473-3315)**

de Dear R (2011) Revisiting an old hypothesis of human thermal perception: alliesthesia. Build Res Inf 39(2):108–117. doi:10.1080/09613218.2011.552269 **(ISSN 0961-3218)**

de Dear R, Brager GS (1998) Developing an adaptive model of thermal comfort and preference. ASHRAE Trans **(ISSN 00012505)**

Echeverría JC, Woolfson MS, Crowe JA, Hayes-Gill BR, Croaker GDH, Vyas H (2003) Interpretation of heart rate variability via detrended fluctuation analysis and alphabeta filter. Chaos 13(2):467–475. doi:10.1063/1.1562051 **(ISSN 1054-1500)**

Fanger PO (1970) Thermal comfort: analysis and applications in environmental engineering. Danish Technical Press, Vanlose (**Republished by McGraw-Hill, New York, 1973**)

Fleisher LA, Frank SM, Sessler DI, Cheng C, Matsukawa T, Vannier CA (1996) Thermoregulation and heart rate variability. Clin Sci Lond 90(2):97–103. doi:10.1042/cs0900097 **(ISSN 0143-5221)**

Fonseca DS, Affonseca Netto AD, Ferreira RB, Miranda De Sa AMFL (2013) Lomb-scargle periodogram applied to heart rate variability study. ISSNIP Biosignals Biorobot Conf BRC 2:8–11. doi:10.1109/BRC.2013.6487524 **(ISSN 23267771)**

Fountain M, Brager G, De Dear R (1996) Expectations of indoor climate control. Energy Build 24(3):179–182. doi:10.1016/S0378-7788(96)00988-7 **(ISSN 03787788)**

Hammel HT, Pierce JB (1968) Regulation of internal body temperature. Annu Rev Physiol 30(1):641–710. doi:10.1146/annurev.ph.30.030168.003233 **(ISSN 0066-4278)**

Hardstone R, Poil SS, Schiavone G, Jansen R, Nikulin VV, Mansvelder HD, Linkenkaer-Hansen K (2012) Detrended fluctuation analysis: a scale-free view on neuronal oscillations. Front Physiol 3:1–13. doi:10.3389/fphys.2012.00450 **(ISSN 1664042X)**

Hoyt T, Arens E, Zhang H (2014) Extending air temperature setpoints: simulated energy savings and design considerations for new and retrofit buildings. Build Environ 88:89–96. doi:10.1016/j.buildenv.2014.09.010 **(ISSN 03601323)**

Huizenga C, Abbaszadeh S, Zagreus L, Arens E (2006) Air quality and thermal comfort in office buildings: results of a large indoor environmental quality survey. Proc Heal Build III:393–397

Humeau-Heurtier A (2015) The multiscale entropy algorithm and its variants: a review. Entropy 17(5):3110–3123. doi:10.3390/e17053110. http://www.mdpi.com/1099-4300/17/5/3110/ **(ISSN 1099-4300)**

Humphreys MA, Hancock M (2007) Do people like to feel 'neutral'?. Exploring the variation of the desired thermal sensation on the ASHRAE scale. Energy Build 39(7):867–874. doi:10.1016/j.enbuild.2007.02.014 **(ISSN 03787788)**

International Facility Management Association (2009) Temperature wars: savings vs. comfort. Technical report. International Facility Management Association, Houston

Jones BW (2002) Capabilities and limitations of thermal models for use in thermal comfort standards. Energy Build 34(6):653–659. doi:10.1016/S0378-7788(02)00016-6 **(ISSN 03787788)**

Karjalainen S (2012) Thermal comfort and gender: a literature review. Indoor Air 22(2):96–109. doi:10.1111/j.1600-0668.2011.00747.x **(ISSN 09056947)**

Karjalainen S (2007) Gender differences in thermal comfort and use of thermostats in everyday thermal environments. Build Environ 42(4):1594–1603. doi:10.1016/j.buildenv.2006.01.009 **(ISSN 03601323)**

Kotsiantis SB, Zaharakis ID, Pintelas PE (2006) Machine learning: a review of classification and combining techniques. Artif Intell Rev 26(3):159–190. doi:10.1007/s10462-007-9052-3 **(ISSN 02692821)**

Laguna P, Moody GB, Mark RG (1998) Power spectral density of unevenly sampled data by least-square analysis: performance and application to heart rate signals. IEEE Trans Biomed Eng 45(6):698–715. doi:10.1109/10.678605 **(ISSN 00189294)**

Lake DE, Richman JS, Griffin MP, Moorman JR (2002) Sample entropy analysis of neonatal heart rate variability. Am J Physiol Regul Integr Comp Physiol 283(3):R789–R797. doi:10.1152/ajpregu.00069.2002 **(ISSN 0363-6119)**

Lin TP, de Dear R, Hwang RL (2011) Effect of thermal adaptation on seasonal outdoor thermal comfort. Int J Climatol 31(2):302–312. doi:10.1002/joc.2120 **(ISSN 08998418)**

Liu W, Lian Z, Liu Y (2008) Heart rate variability at different thermal comfort levels. Eur J Appl Physiol 103(3):361–366. doi:10.1007/s00421-008-0718-6 **(ISSN 14396319)**

Lomb NR (1976) Least-squares frequency analysis of unequally spaced data. Astrophys Space Sci 39(2):447–462. doi:10.1007/BF00648343 **(ISSN 0004640X)**

Maestri R, Pinna GD, Porta A, Balocchi R, Sassi R, Signorini MG, Dudziak M, Raczak G (2007) Assessing nonlinear properties of heart rate variability from short-term recordings: are these measurements reliable? Physiol Meas 28(9):1067–1077. doi:10.1088/0967-3334/28/9/008 **(ISSN 0967-3334)**

Mahdavi A, Kumar S (1996) Implications of indoor climate control for comfort, energy and environment. Energy Build 24(3):167–177. doi:10.1016/S0378-7788(96)00975-9 **(ISSN 03787788)**

Meinshausen N, Bühlmann P (2010) Stability selection. J R Stat Soc Ser B Stat Methodol 72(4):417–473. doi:10.1111/j.1467-9868.2010.00740.x **(ISSN 13697412)**

Milicević G (2005) Low to high frequency ratio of heart rate variability spectra fails to describe sympatho-vagal balance in cardiac patients. Coll Antropol 29(1):295–300 **(ISSN 0350-6134)**

Moody GB (1993) Spectral analysis of heart rate without resampling. Comput Cardiol Proc 1:7–10

Nastase I, Croitoru C, Lungu C (2016) A questioning of the thermal sensation vote index based on questionnaire survey for real working environments. Energy Proced 85:366–374 **(ISSN 18766102)**

Natsume K, Ogawa T, Sugenoya J, Ohnishi N, Imai K (1992) Preferred ambient temperature for old and young men in summer and winter. Int J Biometeorol 36(1):1–4. doi:10.1007/BF01208726 **(ISSN 0020-7128)**

Nicol JF, Roaf S (2017) Rethinking thermal comfort. Build Res Inf Table 1:1–5. doi:10.1080/09613218.2017.1301698 **(ISSN 0961-3218)**

Fanger PO (2001) Human requirements in future air-conditioned environments. Int J Refrig 24(2):148–153. doi:10.1016/S0140-7007(00)00011-6 **(ISSN 01407007)**

Pan J, Tompkins WJ (1985) A real-time QRS detection algorithm. Biomed Eng IEEE Trans 32(3):230–236. doi:10.1109/TBME.1985.325532

Parkinson T, de Dear R (2015) Thermal pleasure in built environments: physiology of alliesthesia. Build Res Inf 43(3):288–301. doi:10.1080/09613218.2016.1140932 **(ISSN 0961-3218)**

Parkinson T, de Dear R (2016) Thermal pleasure in built environments: spatial alliesthesia from contact heating. Build Res Inf 44(3):248–262. doi:10.1080/09613218.2016.1140932 **(ISSN 0961-3218)**

Parkinson T, de Dear R (2017) Thermal pleasure in built environments: spatial alliesthesia from air movement. Build Res Inf 45(3):320–335. doi:10.1080/09613218.2016.1140932 **(ISSN 0961-3218)**

Parkinson T, de Dear R, Candido C (2016) Thermal pleasure in built environments: alliesthesia in different thermoregulatory zones. Build Res Inf 44(1):20–33. doi:10.1080/09613218.2015.1059653 **(ISSN 0961-3218)**

Peng CK, Havlin S, Stanley HE, Goldberger AL (1995) Quantification of scaling exponents and crossover phenomena in nonstationary heartbeat time series. Chaos Interdiscip J Nonlinear Sci 5(1):82–87

Penzel T, Kantelhardt JW, Grote L, Peter JH, Bunde A (2003) Comparison of detrended fluctuation analysis and spectral analysis for heart rate variability in sleep and sleep apnea. IEEE Trans Biomed Eng 50(10):1143–1151. doi:10.1109/TBME.2003.817636 **(ISSN 0018-9294)**

Piskorski J, Guzik P (2007) Geometry of the Poincaré plot of RR intervals and its asymmetry in healthy adults. Physiol Meas 28:287–300. doi:10.1088/0967-3334/28/3/005 **(ISSN 0967-3334)**

Richman JS, Moorman JR (2000) Physiological time-series analysis using approximate entropy and sample entropy. Am J Physiol Heart Circ Physiol 278(6):H2039–H2049. doi:10.1103/PhysRevA.29.975 **(ISSN 0363-6135)**

Riganello F, Garbarino S, Sannita WG (2012) Heart rate variability, homeostasis, and brain function. J Psychophysiol 26(4):178–203. doi:10.1027/0269-8803/a000080 **(ISSN 0269-8803)**

Robinson BF, Epstein SE, Beiser GD, Braunwald E (1966) Control of heart rate by the autonomic nervous system. Studies in man on the interrelation between baroreceptor mechanisms and exercise. Circ Res 19(2):400–411 **(ISSN 0009-7330)**

Rodriguez E, Echeverria JC, Alvarez-Ramirez J (2007) Detrended fluctuation analysis of heart intrabeat dynamics. Phys A Stat Mech Appl 384(2):429–438. doi:10.1016/j.physa.2007.05.022 **(ISSN 03784371)**

Sassi R, Cerutti S, Lombardi F, Malik M, Huikuri HV, Peng CK, Schmidt G, Yamamoto Y (2015) Advances in heart rate variability signal analysis: joint position statement by the e-Cardiology ESC Working Group and the European Heart Rhythm Association co-endorsed by the Asia Pacific Heart Rhythm Society. Europace 17(9):1341–1353. doi:10.1093/europace/euv015 **(ISSN 15322092)**

Schiavon S, Hoyt T, Piccioli A (2014) Web application for thermal comfort visualization and calculation according to ASHRAE Standard 55. Build Simul 7(4):321–334. doi:10.1007/s12273-013-0162-3 **(ISSN 1996-3599)**

Selvaraj N, Jaryal A, Santhosh J, Deepak KK, Anand S (2008) Assessment of heart rate variability derived from finger-tip photoplethysmography as compared to electrocardiography. J Med Eng Technol 32(6):479–484. doi:10.1080/03091900701781317 **(ISSN 0309-1902)**

Shannon CE (1948) A mathematical theory of communication. Bell Syst Tech J 27(3):379–423. doi:10.1002/j.1538-7305.1948.tb01338.x. http://cm.bell-labs.com/cm/ms/what/shannonday/shannon1948.pdf, http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=6773024 **(ISSN 00058580)**

Sunkaria RK (2011) Recent trends in nonlinear methods of HRV analysis: a review. Eng Technol 75(3):566–571 **(ISSN 2010376X)**

Task Force of The European Society of Cardiology and The North American Society of Pacing and Electrophysiology (1996) Heart rate variability: standards of measurement, physiological interpretation, and clinical use. Circulation 93(5):1043–1065

Thayer JF, Nabors-Oberg R, Sollers JJ (1997) Thermoregulation and cardiac variability: a time-frequency analysis. Biomed Sci Instrum 34:252–256 **(ISSN 0067-8856)**

Van Hoof J (2008) Forty years of Fanger's model of thermal comfort: comfort for all? Indoor Air 18(3):182. doi:10.1111/j.1600-0668.2007.00516.x **(ISSN 09056947)**

Voss A, Heitmann A, Schroeder R, Peters A, Perz S (2012) Short-term heart rate variability—age dependence in healthy subjects. Physiol Meas 33(8):1289–1311. doi:10.1088/0967-3334/33/8/1289 **(ISSN 09673334)**

West BJ (2010) Fractal physiology and the fractional calculus: a perspective. Front Physiol:1–17. doi:10.3389/fphys.2010.00012 **(ISSN 1664042X)**

Willson K, Francis DP, Wensel R, Coats AJ, Parker KH (2002) Relationship between detrended fluctuation analysis and spectral analysis of heart-rate variability. Physiol Meas 23(2):385–401. doi:10.1088/0967-3334/23/2/314 **(ISSN 0967-3334)**

Yao Y, Lian Z, Liu W, Jiang C, Liu Y, Lu H (2009) Heart rate variation and electroencephalograph—the potential physiological factors for thermal comfort study. Indoor Air 19(2):93–101. doi:10.1111/j.1600-0668.2008.00565.x **(ISSN 09056947)**

Ye XJ, Zhou ZP, Lian ZW, Liu HM, Li CZ, Liu YM (2006) Field study of a thermal environment and adaptive model in Shanghai. Indoor Air 16(4):320–326. doi:10.1111/j.1600-0668.2006.00434.x **(ISSN 0905-6947)**

Yentes JM, Hunt N, Schmid KK, Kaipust JP, McGrath D, Stergiou N (2013) The appropriate use of approximate entropy and sample entropy with short data sets. Ann Biomed Eng 41(2):349–365. doi:10.1007/s10439-012-0668-3 **(ISSN 1573-9686)**